Report Biomimicry Challenge

ZHAW Institute for Sustainable Development | Blockweek Bionics 2026

Challenge 1 — Urban Heat Islands

Cooling the City

"Picture"

Insert a concept image here:
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Students	**Contribution to the report / (sections)**
[TODO: Name 1]	[TODO: e.g. Sections 2, 3.2, engineering calculations]
[TODO: Name 2]	[TODO: e.g. Sections 3.1, 4, biological research]
[TODO: Name 3]	[TODO: e.g. Sections 1, 5, figures and editing]
[TODO: Name 4]	[TODO: e.g. Section 3.3, measurements, appendix]

Comments	[TODO: optional remarks for the instructors]

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Contents

1 Introduction

1.1 Background and relevance

[TODO: 2–3 paragraphs. Suggested content, to be written out and referenced:]

- Urban heat islands (UHI): built-up areas are warmer than their rural surroundings, mainly because impervious surfaces absorb short-wave radiation, store the heat and release it at night, while evapotranspiration and long-wave loss to the sky are reduced (Oke 1982).
- Magnitude in the reference city: Zurich reports an urban heat island of 1 K to 2 K by day and up to 7 K at night in the centre relative to the surrounding area, and sealed sun-exposed surfaces exceeding 60 °C in summer (Stadt Zürich 2021). The night-time load is the larger share, and it is the share that impairs sleep and recovery.
- Health relevance: the city names cerebrovascular, cardiovascular and respiratory illness during heat periods, with older people and small children particularly at risk (Stadt Zürich 2021). Mechanical cooling would raise both electricity demand and the waste heat released into the street canyon, which is why the city advises against air conditioning (Stadt Zürich 2020a, pp. 132–133).
- Policy context: Zurich's *Fachplanung Hitzeminderung* (Stadt Zürich 2020a) and its implementation agenda (Stadt Zürich 2020c) make retrofit measures on existing building envelopes an actively pursued field of application — and define the incumbent solution this project has to improve on (Section 2.3).

1.2 Aim of the report and intended contribution

[TODO: State the aim in 3–5 sentences, e.g.:] This report documents a biomimicry design process for reducing heat load on the envelope of existing urban buildings. It follows the Biomimicry Design Spiral (Baumeister et al. 2014) through scoping, biological discovery and abstraction, engineering concept development, validation and evaluation. The intended contribution is a technically specified, quantitatively checked envelope concept, together with a transparent account of which biological mechanisms were transferred, which were rejected, and on what evidence.

1.3 Structure of the report

[TODO: One short paragraph describing Sections 2–5.]

2 Challenge & Scoping

2.1 Problem definition and design question

Given challenge (Challenge 1). *Cities are facing increasing heat stress due to climate change. Develop bioinspired strategies to reduce urban heat through building materials, façade design, or urban planning.*

The challenge as given spans three scales — material, building component and urban planning. Working on all three would make the project larger rather than more focused, which the systems view is explicitly meant to prevent. The design question is the instrument that makes this cut.

2.1.1 Candidate questions considered

Five candidates were formulated in the team and assessed against the criterion that a design question must be neither too broad nor too narrow: specific enough to define an intervention point, open enough to admit different solutions.

Table 1: Design questions considered, and why four were rejected.

#	Candidate	Assessment
1	How do we prevent heat build-up in urban areas?	*Too broad.* No place, no time, no intervention point — the equivalent of "How might we end hunger?"
2	How could we use the energy or heat build-up positively while lowering temperature?	Right idea, but unanchored: no location, and "positively" does not say for what. Refined into candidate 5.
3	How might we get the heat that Zurich's existing buildings store during the day back out of the city before the next morning, without using energy?	Well formed and highly testable. Rejected only because it discards the team's central idea of *using* the heat. Its night-time focus is retained in the selected question.
4	How can we prevent heat waves causing heat islands?	*Factually wrong.* Heat waves do not cause heat islands. The urban heat island is a permanent consequence of urban surface properties and geometry and exists on ordinary summer days; a heat wave makes it dangerous, it does not create it (Oke 1982).
5	How might we turn the heat that accumulates in a city into a useful resource, instead of reflecting or discarding it?	Good direction, but the reading "store summer heat for winter" is not verifiable within the project, and it has no biological support: organisms store chemical energy across seasons, not sensible heat. Narrowed into the selected question.

2.1.2 Selected design question

Design question (solution-neutral):

How might we let the heat that Zurich's existing buildings absorb during a summer day drive its own removal, so that the hotter it gets, the more the building cools itself?

The question names no solution — no coating, no greening, no geometry, no water. It states the location (Zurich), the object (existing buildings, so retrofit rather than new build) and the operating season. What it adds to candidates 3 and 5 is the subordinate clause, and that clause is the actual content of the project: it demands a **self-regulating** response, in which the cooling power rises with the heat load instead of being fixed.

This matters because it distinguishes the project from the incumbent measure. A bright surface (Section 2.3) has a fixed optical response: it reflects the same fraction of radiation at 20 °C as at 40 °C, and it does nothing at all once the sun has set. A self-regulating surface behaves like a biological one — transpiration in a leaf is driven by the same solar energy that heats the leaf, so the cooling effort scales with the load, without a controller, a sensor or a pump.

It also resolves the team's initial difficulty of having two systems of interest ("reduce heat" and "prevent heat"). These are not two systems. They are two functions of one system, coupled by the selected question: prevention acts during the day window, removal during the night window, and the self-regulating link connects them.

2.2 Context and constraints

Reference city: Zurich. All context values, benchmarks and engineering checks in this report refer to the City of Zurich (47.38° N, 8.54° E, approximately 400 m above sea level on the valley floor of the Limmat). Fixing one reference city is what makes the constraints below and the KPI targets in Table 8 checkable instead of generic. Zurich was chosen for four reasons:

1. The city has an adopted municipal heat-mitigation strategy, the *Fachplanung Hitzeminderung* (Stadt Zürich 2020a), with an implementation agenda (Stadt Zürich 2020c), a guidance leaflet for building owners (Stadt Zürich 2021) and three thematic maps (Stadt Zürich 2020b). The project therefore has a real client, a documented baseline and published target values instead of invented ones. Section 2.3 shows what this means for our design space.
2. The city publishes an open climate analysis and open geodata (Stadt Zürich n.d.), so the reference site can be selected from measured heat-load data instead of intuition.
3. MeteoSwiss operates long-term stations in and around the city (MeteoSwiss n.d.), which supplies the boundary conditions (G, T_a, wind speed, humidity) for the energy balance in Section 3.2.
4. Its building stock and morphology are representative of the Swiss Mittelland, so results transfer to comparable Swiss cities.

Reference site and users. We adopt the city's own model area *Modellierungsgebiet 03 — Geschlossene Randbebauung* (closed perimeter block) as the reference site (Stadt Zürich 2020a, pp. 148–151). It consists of two outward-closed blocks of five storeys plus a roof structure enclosing private courtyards, in a dense district close to the centre; the street space is sealed from façade to façade and only isolated trees provide shade. The city's own assessment is that the reduced air exchange and the lack of cold-air supply make this structural type overheat comparatively quickly.

Choosing this site has a methodical advantage: the Fachplanung already publishes modelled baseline and climate-optimised results for exactly this area, so our concept can be compared against a documented reference rather than against our own assumptions. **[TODO: Record**

address, façade orientations, roof area and year of construction of the specific building you pick inside this area.]

The users are the residents of the upper floors, who carry the strongest heat load and usually have no mechanical cooling; the pedestrians in the street canyon, who receive radiation from the façade at eye level; the building owner, who decides on and pays for the retrofit; and the trades who install and later maintain the system.

When — the design case. The design case is not a single hot afternoon but a **multi-day heat wave**: five consecutive hot days ($T_{max} \geq 30\,°C$) without rain. This matters because the concept relies on a stored resource. A system that performs well on day one and is exhausted on day three has not solved the problem, and only a multi-day case puts the water reserve (F7) and the reset behaviour of the heat store (F4) under real stress. The autonomy requirement in KPI 6 follows from this case. **[TODO: Confirm the duration and the frequency of such spells for Zurich from MeteoSwiss records, and state the station and reference period.]**

When — the critical windows. Within each day of that heat wave the design must perform in three windows, which follow one another without a gap because the surface never stops exchanging heat. Sunset in Zurich in high summer is around 21:00, sunrise around 06:00.

Table 2: Critical operating windows within one day of the design heat wave.

Window	What dominates	Why it is critical	KPIs
Load 11:00–18:00	Absorbed short-wave radiation. Clear-sky global irradiance of the order of 900 W/m^2 to 950 W/m^2 on a horizontal surface around solar noon (about 13:30 local time), air temperature up to roughly 33 °C to 36 °C.	The energy that has to be dealt with all night enters here. A sealed, sun-exposed surface can exceed 60 °C (Stadt Zürich 2021). The city evaluates PET at 14:00, which lies inside this window, so our results stay comparable to Table 4.	2, 3, 4
Evening 18:00–24:00	Until sunset, low-angle sun on west-facing surfaces; after sunset, the start of the discharge of stored heat into the street canyon.	This is when residents are outdoors and then go to bed, so it is the window in which the effect is actually experienced. West façades receive their highest incidence angle here, which is why the window starts at 18:00 rather than at 20:00 — the two hours in between are among the hottest of the day at the surface, not a gap.	4, 5
Night 00:00–06:00	Long-wave emission to the sky and continued release of stored heat. Clearest sky conditions of the cycle.	Zurich's urban heat island reaches 7 K at night against 1 K to 2 K by day (Stadt Zürich 2021), so this is the larger half of the problem and the half the incumbent measure does not address. It also sets a hard deadline: the heat store must be discharged and the surface reset by sunrise, or it saturates over the following days of the heat wave.	1, 5, 6

The remaining hours, 06:00–11:00, are not critical: the load is rising but still moderate. They are the margin the night window has to leave behind, and the state of the system at 06:00 is therefore reported as the reset condition for the next day.

Why the night window was extended. The team's initial scoping put the second window at 20:00–24:00. Two things argued for changing it. First, the night-time reference instant of KPI 5 is 02:00, which the original window did not contain — a window must include the moments at which its own KPIs are evaluated. Second, tropical nights are defined over the whole night and the minimum temperature occurs shortly before sunrise, so a window ending at midnight would exclude the part of the night that defines the problem. Splitting the evening from the night also separates two genuinely different concerns: outdoor amenity while people are awake, and sleep quality plus the reset of the store afterwards. They need different KPIs, so they are treated as two windows.

Where on the envelope? A consequence of Zurich's latitude. Fixing the reference city fixes the solar geometry, and that already narrows the design space. At 47.38° N the solar altitude at noon on 21 June is

$$h = 90^\circ - 47.38^\circ + 23.44^\circ \approx 66^\circ. \tag{1}$$

The direct beam therefore strikes a horizontal roof at an incidence factor of $\sin h \approx 0.91$, but a vertical south-facing façade at only $\cos h \approx 0.41$. In midsummer the flat roof receives more than twice the direct irradiance of a south façade at noon, and it also has a far larger sky view factor for night-time long-wave emission. The system of interest is therefore deliberately *not* restricted to the façade. **[TODO: Confirm with daily irradiation totals per orientation (Swiss Federal Office of Energy n.d.), then justify whether the concept targets roof, façade or both. East and west façades peak in the morning and late afternoon, so a façade concept has a different load profile, not simply a smaller one.]**

Constraints. Table 3 separates hard constraints (a violation disqualifies a concept) from soft ones (they weight the evaluation in Section 3.3). The constraints on installation and maintenance come from the team's observation that the people who mount and service the system are part of the context; they are recorded here rather than as subsystems, because the system of interest contains components, not people.

2.3 State of the art: what the City of Zurich already does

Scoping is incomplete without knowing what the client already does. The Fachplanung Hitzeminderung defines thirteen courses of action (*Handlungsansätze*, HA 01–HA 13). Four act on the building envelope and therefore touch our project: HA 09 green roofs, HA 10 green façades, HA 11 high-albedo façade and roof materials, and HA 12 shading of the outdoor space close to the building (Stadt Zürich 2020a, pp. 120–131). HA 11 is the direct incumbent of our design question, and the city already recommends it to building owners (Stadt Zürich 2021).

Baseline values this fixes for us. The city's model uses a solar reflectance of 0.2 for asphalt and 0.3 for un-greened façades, and classifies ≤ 0.30 as low and ≥ 0.80 as high albedo. Reported material values are slate 0.1, roof tiles 0.1–0.35, natural stone 0.2–0.35 and a white façade 0.5–0.9 (Stadt Zürich 2020a, pp. 114, 128–129). Our reference case and the target of KPI 2 in Table 8 are taken directly from these figures, so the comparison is made on the client's terms rather than on ours.

The gap — and the design opportunity. Table 4 contains the central finding for this project. The incumbent measure works during the day but is close to ineffective at night: −0.1 K median air temperature, against an urban heat island that reaches 7 K at night. The reason is structural, not a question of tuning. A bright surface reduces what is *absorbed*; it provides no

Table 3: Context conditions and constraints.

Type	Class	Constraint
Energy	hard	No auxiliary electrical energy for the cooling function itself. This follows directly from the design question: the heat load must drive the response. Sensors used for validation are excluded.
Water	hard	No potable water. Only rainwater harvested on site or atmospheric moisture.
Existing building	hard	Retrofit onto an existing envelope. Load-bearing structure, floor plan and window openings are not redesigned.
Townscape	hard	The result must be acceptable in the Zurich townscape. Buildings may be listed; heritage protection, building law and neighbour rights can restrict visible changes (Stadt Zürich 2020a, p. 128).
Glare	hard	No specular glare towards streets, neighbouring flats or traffic. The city names glare as the main objection to bright envelopes (Stadt Zürich 2020a, p. 128), which favours diffuse over mirror-like surfaces.
Climate	hard	Must survive the Swiss annual cycle: frost to about −10 °C, freeze–thaw cycles, driving rain, UV, and periods of several dry weeks in summer.
Winter	hard	Must not appreciably increase heating demand. A high solar reflectance also reduces useful winter solar gains; this trade-off must be quantified, not ignored.
Installation	soft	Must be installable by ordinary trades on an occupied building, ideally without full scaffolding. **[TODO: state an installation-time target per m^2]**
Maintenance	soft	Must remain functional with little maintenance. Soiling and the resulting cleaning cost are named by the city as a specific weakness of high-albedo envelopes (Stadt Zürich 2020a, p. 128). Service life target ≥ 20 years.
Solar energy	soft	Should not unnecessarily compromise roof- or façade-integrated photovoltaics; the city notes that bright envelopes reduce PV yield (Stadt Zürich 2020a, p. 128). State which envelope areas the concept claims.
Structural	soft	Limited additional dead load and build-up thickness on an existing envelope.
Economic	soft	The city expects higher cost for high-albedo envelopes (Stadt Zürich 2020a, p. 128). Target: cost per square metre in the range of a normal refurbishment. **[TODO: find a benchmark figure and cite it]**

Table 4: Modelled effectiveness of the incumbent measures, from the Fachplanung Hitzeminderung. Day values are the physiologically equivalent temperature (PET) at 14:00, 2 m above ground; night values are air temperature. "Range" is the distance over which the effect is measurable.

Measure	Median	Max.	Range	Source
HA 11 bright façade — day (PET)	−1.0 K	−1.8 K	2 m to 3 m	(Stadt Zürich 2020a, Tab. 11, 128)
HA 11 bright façade — night	−0.1 K	−0.2 K	1 m to 2 m	
HA 11 bright roof — day (PET)	−1.2 K	−2.1 K	2 m to 4 m	
HA 11 bright roof — night	−0.1 K	−0.3 K	1 m to 2 m	
HA 06 high-albedo ground surface instead of asphalt — day (PET)	−1.5 K	−2.8 K	2 m to 4 m	(Stadt Zürich 2020a, Tab. 5, 114)
HA 06 — night	−0.2 K	−0.5 K	2 m to 3 m	

path for releasing heat that is already stored in the construction, and its optical response is fixed — it does not do more when the load is higher, and it does nothing after sunset.

This is exactly what the selected design question targets. A bio-inspired concept earns its place here if it does at least one of the following:

1. deliver a night-time effect that HA 11 does not, by opening a removal path rather than only a prevention path;
2. scale its response with the load, so that the worst hours receive the strongest cooling;
3. resolve one of the constraints the city itself names for HA 11 — above all glare and soiling, which are precisely the points where biological surfaces are strong.

[TODO: Note for Section 5: the running text on p. 128 of the Fachplanung quotes a maximum PET reduction of 2.3 K for a south-west facing façade, while Tab. 11 on the same page gives 1.8 K. Check the source, decide which value you use, and say so. Spotting this is worth more than quietly copying one of the two.]

2.4 System view and project boundary

Figure 1 places the challenge in its system hierarchy. The hierarchy prevents tunnel vision; the boundary drawn afterwards prevents scope explosion.

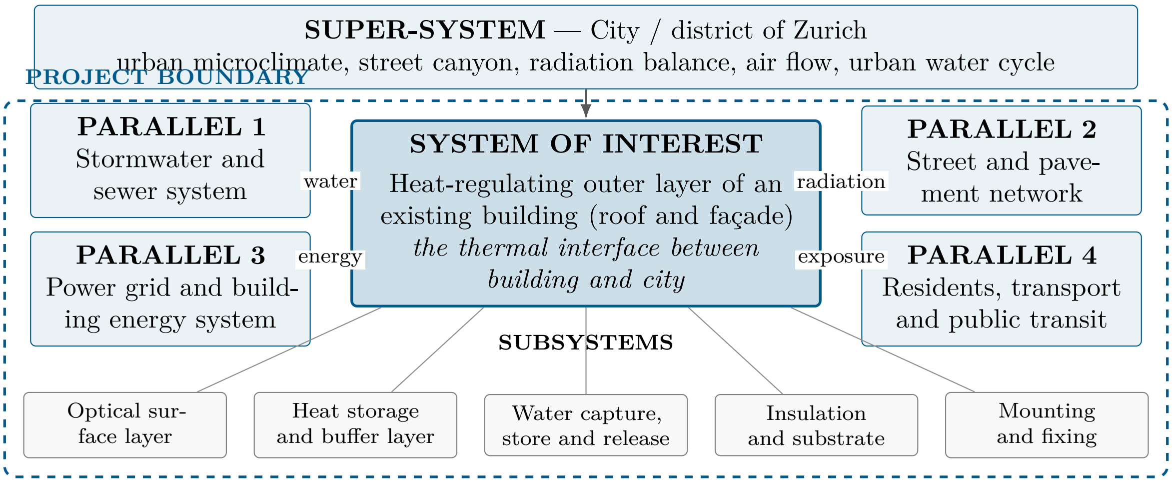

Figure 1: System view. The dashed line is the project boundary: everything inside it is designed and analysed by the team. Note that the subsystems are components and functions, not people — the trades who install and maintain the system appear as constraints in Table 3.

What each parallel system adds to the scoping. Listing parallel systems is not the point; what the comparison reveals is.

What the systems view changed. *The systems view changed our project scope because* it forced two corrections. First, we initially carried two systems of interest — reducing heat and preventing heat. The hierarchy showed that these are two *functions* of one system, and the selected design question couples them, so a single boundary can be drawn. Second, the comparison with the parallel systems revealed the night sky and the discarded rainwater as the two heat sinks available without energy input, and the pedestrian in the street canyon as the place where the effect must actually appear. This added the functions F2, F6–F8, the evening operating window, and KPI 5 and 6 in Table 8.

Table 5: What the comparison with each parallel system contributes.

Parallel system	What the comparison adds
Stormwater and sewer system	Rainwater is currently drained away as fast as possible. Seen from our system it is a cooling reserve that is being discarded — and during a heat wave it is the only water available under the "no potable water" constraint. This adds the functions *capture*, *retain* and *meter* water (F6–F8) and makes storage capacity a design variable rather than an afterthought.
Street and pavement network	The street is treated by the same client with the same logic (HA 06, Table 4), and it achieves a slightly larger daytime effect than the envelope. The comparison sets a benchmark we must beat, and it shows that the envelope's advantage has to come from the night window, where the street performs no better.
Power grid and building energy system	The obvious way to "use" heat is to convert it. The comparison shows why that is a different project: conversion competes with photovoltaics for the same surface, and any seasonal store lies outside the envelope. It is therefore placed out of the boundary, while the milder reading of "use" — letting the heat drive its own removal — stays in.
Residents, transport and public transit	Defines where the effect has to be delivered: at pedestrian level in the street canyon and in the flats on the upper floors, not on the roof membrane. This is why PET at 2 m above ground becomes a KPI, and it fixes the evening window 20:00–24:00, when people are outdoors and then sleeping.

Table 6: Project boundary: what is deliberately in and out of the project.

IN the project	OUT of the project
Passive control of the short-wave and long-wave radiation balance of the envelope surface	Redesign of the building itself: structure, floor plan, windows, insulation thickness
The self-regulating link between heat load and cooling power — the core of the design question	Any actively controlled system: sensors, valves, pumps, chillers, mechanical ventilation
Heat storage and delayed release within the outermost build-up, including the night window	Seasonal heat storage, boreholes, heat pumps, thermal networks and electricity generation
Capture, retention and metered release of rainwater within the build-up	Municipal water supply, treatment and sewage; street trees and district-scale greening
Mounting on an existing substrate, installability and durability of the function	Permitting, heritage approval, business model, full life cycle assessment

2.5 Functions and functional diagram

Functions are formulated as solution-neutral verbs answering "what must the design *do*?". They are stated at the scale of the designed object, not at the scale of the city: the design regulates heat at the surface it occupies, and the city-scale effect is the *goal*, not the function — in the same way that the function of polar bear fur is to insulate, not to make the Arctic habitable.

Table 7: Function list. Main function, sub-functions and supporting functions.

ID	Function (verb)	Explanation / taxonomy link
Main function		
F0	Regulate heat gain and release at an urban surface	Modify / regulate physical state: manage temperature
Sub-functions — prevention (day window)		
F1	Reflect short-wave radiation	Prevent absorption of solar energy in the first place
F5	Shade a surface	Reduce the irradiated area, e.g. through surface geometry
Sub-functions — removal (both windows)		
F2	Emit long-wave radiation	Release heat to the sky, in particular in the atmospheric window
F3	Transfer heat to ambient air	Convective dissipation at the surface
F4	Store thermal energy temporarily	Buffer and shift the daily peak
Sub-functions — water path		
F6	Capture water	Collect rain or atmospheric moisture
F7	Retain water	Store it until it is needed during the heat peak
F8	Release water in a controlled way	Meter evaporation so the reserve lasts the whole heat wave
Sub-function — the core of the design question		
F9	Couple response strength to load	Increase removal (F2, F3, F8) as surface temperature rises, without external control
Supporting functions		
F10	Keep a surface clean	Maintain the optical function against soiling and algae
F11	Resist degradation	Withstand UV, frost, driving rain over the service life
F12	Attach to a substrate	Fix the build-up to an existing wall or roof, installable by ordinary trades

Why F9 is physically plausible. Every removal path already depends on surface temperature, but not equally strongly. Differentiating the three mechanisms with respect to surface temperature T_s gives their self-regulation gradients, in $\mathrm{W/(m^2\,K)}$:

$$\text{radiative:} \quad \frac{\partial P}{\partial T_s} = 4\varepsilon\sigma T_s^3 \approx 6 \tag{2}$$

$$\text{convective:} \quad \frac{\partial P}{\partial T_s} = h_c \approx 10 \tag{3}$$

$$\text{latent:} \quad \frac{\partial P}{\partial T_s} \propto \frac{\mathrm{d}e_s}{\mathrm{d}T_s} \quad \text{(steep — } e_s \text{ roughly doubles per } 10\,\mathrm{K}) \tag{4}$$

The evaporative path is by far the steepest, which is why transpiration, not radiation, is the mechanism that biology uses when a load must be tracked. **[TODO: Section 3.2: evaluate**

these three derivatives properly with your own assumptions and state the resulting ranking. This is the check that decides which mechanism carries F9.]

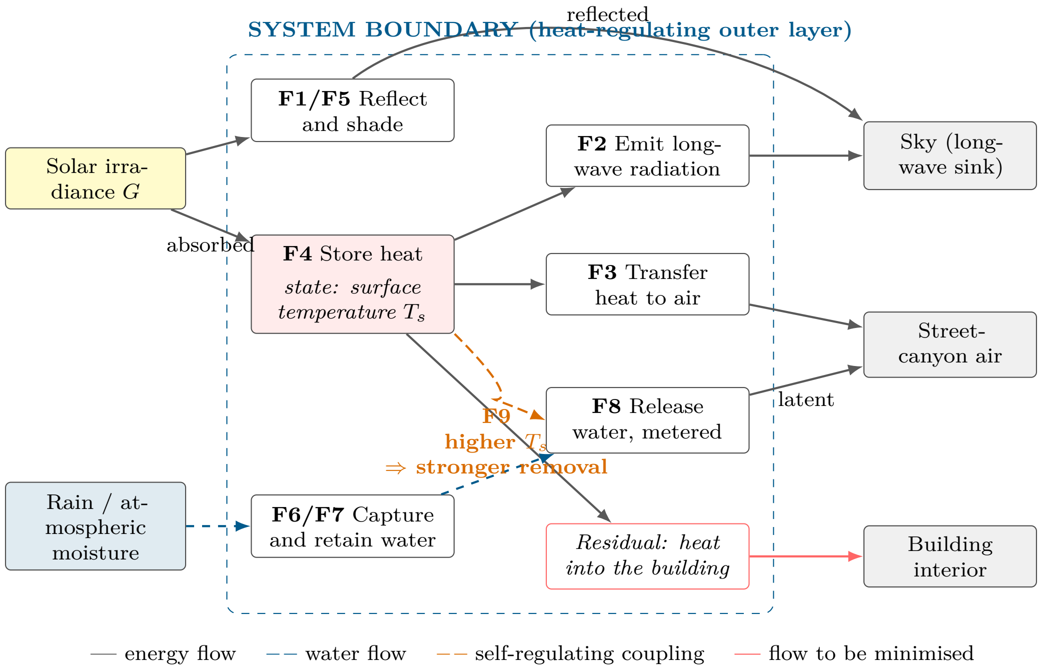

Figure 2: Solution-neutral functional diagram. Only functions and flows are shown, no components and no materials. The orange coupling is F9 and is what distinguishes this project from a fixed high-albedo surface: the state variable T_s feeds back into the strength of the removal functions. Supporting functions F10–F12 act on everything inside the boundary and are omitted for readability.

2.6 Biologized questions

The functions are translated into questions to nature, which form the search strategy for Section 3.1:

- How does nature use the heat it receives to get rid of that same heat?
- How does nature make a response stronger when the load is stronger, without a controller?
- How does nature release excess heat to the night sky?
- How does nature reflect intense solar radiation without pigments that bleach, and without a mirror finish?
- How does nature capture, store and meter out scarce water so that a reserve lasts a whole dry period?
- How does nature buffer a daily temperature extreme instead of resisting it?
- How does nature keep a functional surface clean without cleaning it?

2.7 Performance criteria and KPIs

The KPIs are derived from the functions, interfaces and constraints identified above, not invented independently of them. They are used twice: to guide the engineering checks in Section 3.2 and to compare the concepts in Section 3.3.

The reset condition. KPI 1 and KPI 5 are evaluated per day, but over the five-day design case a further condition applies: the surface temperature and the state of charge of the heat

Table 8: Performance criteria. Function → KPI → unit → target. Reference case: an existing Zurich façade or roof with a solar reflectance of about 0.3, the value the city uses for un-greened façades (Stadt Zürich 2020a, p. 114).

#	Function	KPI	Unit	Target / benchmark
1	F9 Couple response to load	Self-regulation gradient $\partial P_{cool}/\partial T_s$	$W/(m^2\,K)$	Must be clearly positive and larger than for a fixed high-albedo surface. **The defining criterion of this project** — it is the quantity the design question asks for. **[TODO: set a target after the derivative check in Section 3.2]**
2	F1 Reflect short-wave radiation	Solar reflectance α_{sol} (300–2500 nm)	–	≥ 0.80, the threshold the city defines for "high albedo" (Stadt Zürich 2020a, p. 128). Baseline 0.3. Ultra-white laboratory paints reach ≈ 0.98 (Li et al. 2021).
3	F0 Regulate heat	Peak surface temperature reduction ΔT_s vs. reference, clear sky, 14:00	K	≥ 15 K. **[TODO: confirm by calculation or measurement before claiming it]**
4	F0 (day, people)	PET reduction at 2 m above ground, 14:00	K	Beat the incumbent: HA 11 achieves -1.0 K (façade) and -1.2 K (roof) median (Stadt Zürich 2020a, Tab. 11). This is the metric the client uses.
5	F2, F4 (evening / night)	Net cooling power at 02:00, clear sky; and heat discharged over the evening and night windows, 18:00–06:00	W/m^2; $W\,h/m^2$	$\geq 40\ W/m^2$ instantaneous (Raman et al. 2014; Zhai et al. 2017), and a night-time air temperature effect better than the -0.1 K of HA 11 (Stadt Zürich 2020a, Tab. 11).
6	F6–F8 Water path	Potable water consumption; and days of autonomy on harvested rainwater	$L/(m^2\,d)$; d	Potable water = 0 (hard constraint). Autonomy ≥ 5 days, so the reserve survives the five-day design heat wave of Table 2. **[TODO: derive the required storage from the latent heat of vaporisation, 2.45 MJ/kg]**
7	F10/F11 Durability	Retention of the cooling function after soiling and weathering	% of initial	≥ 90 % after three years. Soiling is the weakness the city names for high-albedo envelopes (Stadt Zürich 2020a, p. 128); aged cool roofs commonly lose a significant fraction of their initial reflectance (Santamouris 2014).
8	Constraints: economy and effort	Installed cost; and embodied greenhouse gas of the added build-up	CHF $1/m^2$; kg CO_2-eq $1/m^2$	Within the range of a conventional refurbishment, and the embodied emissions must be repaid by the avoided cooling demand within the service life. **[TODO: both benchmarks still to be sourced]**

store at 06:00 must not drift upwards from one day to the next. A concept that cools well on the first day but saturates by the third fails the design case even if every single-day KPI is met. **[TODO: report the five-day trend in Section 3.2, not only day one]**

Two changes to the team's original KPI list. Two of the KPIs originally proposed were reformulated, in both cases because of the rule that KPIs must follow from the functions and the system view.

Urban heat island intensity was dropped as a selection KPI. It is a district-scale quantity, and a single retrofitted building cannot move it measurably; keeping it would invite exactly the confusion between surface and air temperature described below. It remains in Section 1 as the context that motivates the project, and it is replaced operationally by KPI 4, which measures the same concern where it is actually attributable to our system.

Sustainability cost was not a measurable quantity as stated. It is replaced by the embodied-emissions half of KPI 8, which has a unit, a benchmark and a decision rule.

One quantity deliberately not a KPI. The heat conducted through the envelope into the building is reported in Section 3.2 as a secondary result, because it follows from the same energy balance at no extra cost and matters to the building owner. It is not a selection KPI: indoor comfort and building services lie outside the project boundary (Table 6).

Important distinction. KPI 3 measures a *surface* temperature, KPI 4 an effect on people. A high-reflectance surface can be 20 K cooler than a dark one while the effect on the air temperature of the district is smaller by roughly two orders of magnitude (Santamouris 2014; Akbari et al. 2009). Conflating the two is the most common error in urban-heat-island projects, so the two are reported separately throughout this report.

Note on the targets. The targets above are aspirational benchmarks set during scoping. They are hypotheses, not results. Each is checked in Section 3.2 and either confirmed, revised or explicitly declared unverified. **[TODO: before hand-in: verify every cited benchmark value against its primary source]**

2.8 Relevant Life's Principles

The Life's Principles most relevant to this challenge, used again as a sustainability benchmark in Section 4:

- **Use low energy processes** — the whole cooling function must be passive. This is the hard energy constraint and follows from the design question itself.
- **Be locally attuned and responsive** — the principle that the design question is built on. The response must track the local load and the locally available water, not an annual average. F9 is this principle expressed as a function.
- **Adapt to changing conditions** — performance is required in a heat wave, in an ordinary summer and in winter, from one surface.
- **Be resource efficient (material and energy)** — fit form to function; low added mass and low embodied emissions per square metre (KPI 8).
- **Maintain integrity through self-renewal** — the function degrades through soiling; a self-cleaning surface preserves KPI 7 over the service life.
- **Use life-friendly chemistry** — relevant when selecting pigments, binders and any biocide in the outer layer.

Quality check.

- ☐ Is our system of interest clearly defined? — Yes, and singular: the heat-regulating outer layer of an existing building.
- ☐ Is our project boundary manageable? — Yes, see Table 6. Seasonal storage and electricity generation are explicitly out.
- ☐ Are our functions solution-neutral? — Yes, verbs only; no components in Table 7 or Figure 2.
- ☐ Did the systems view reveal additional functions or constraints? — Yes: F2 and F6–F8 from the sky and the discarded rainwater, the evening window from the residents, and the installation and maintenance constraints.
- ☐ Are our most important functions measurable? — Yes, KPIs 1–8, each with a unit.
- ☐ Is the "How might we. . . ?" question open to different solutions? — Yes: evaporative, radiative and geometric strategies all remain possible, and Section 3.2 must choose between them on evidence.

3 Bio-Inspired Design Process

[TODO: One short paragraph: how the team organised the search and the design path, so the reasoning chain from Section 2 to Section 3.3 is visible.]

3.1 Discovering & Abstracting

3.1.1 Research path

[TODO: Describe how you searched: AskNature functions used, search terms, databases, which biologized question from Section 2.6 led to which model.]

Table 9: Biological models found, and the selection / de-selection decision.

Organism / model	Function	Mechanism (short)	Selected?
Saharan silver ant *Cataglyphis bombycina*	F1, F2	Triangular hairs scatter visible and near-infrared light and simultaneously enhance emissivity in the mid-infrared (Shi et al. 2015)	**[TODO: yes/no + why]**
White beetle *Cyphochilus*	F1	Brilliant whiteness from a disordered chitin network only a few micrometres thick, without pigment (Vukusic et al. 2007)	**[TODO: yes/no + why]**
African elephant skin	F6–F8	Network of micrometre-scale cracks retains water and extends evaporative cooling (Martins et al. 2018)	**[TODO: yes/no + why]**
Desert shrub *Encelia farinosa*	F1, F5	Dense leaf pubescence raises reflectance and is adjusted seasonally (Ehleringer and Mooney 1978)	**[TODO: yes/no + why]**
Termite mound	F3	Classic biomimicry reference, but the "chimney air-conditioning" account is contested; measurements support wind-driven transient gas exchange rather than a thermosiphon (Turner and Soar 2008)	**[TODO: recommended: DE-SELECT with this justification]**
[TODO: further models]			

3.1.2 Mechanisms in depth

[TODO: For the 2–3 selected models: explain the mechanism properly — geometry, length scales, physics, and the evidence. Add a sketch. This is worth 20 % of the grade and descriptive text alone is not enough.]

3.1.3 Abstraction into design principles

[TODO: Fill in. Keep the three columns strictly separate.]

Table 10: From biological observation to transferable design principle.

Biological observation	Abstracted principle	Possible technical interpretation
[TODO: what was observed]	[TODO: the principle, free of the organism]	[TODO: how it could be built — not yet a decision]

3.2 Creating: Engineering Concept Development

3.2.1 Ideation

[TODO: Document the method (Crazy 8s / morphological matrix). Include a photo or scan of the sheets. Show at least three initial combinations.]

3.2.2 Concept A — [TODO: name]

[TODO: Annotated drawing + description: components, functional chain, flows, candidate materials, key technical parameters.]

3.2.3 Concept B — [TODO: name]

[TODO: Must be technically DISTINCT from A — a different physical principle, not a variant. E.g. a radiative (dry) concept vs. an evaporative or geometric (self-shading) concept.]

3.2.4 Engineering Canvas and critical questions

[TODO: Insert the Engineering Canvas. Then derive 2–3 critical engineering questions of the form: "What must be true for the concept to work?"]

1. **[TODO: Critical question 1]**
2. **[TODO: Critical question 2]**
3. **[TODO: Critical question 3]**

3.2.5 Engineering check 1: steady-state surface energy balance

[TODO: This check is prepared because it is the most direct way to test KPI 1 and KPI 3. Fill in your own numbers, state every assumption, and keep the units.]

For an opaque surface in steady state, the energy balance per unit area is

$$(1-\alpha_{\text{sol}})\,G \;+\; \varepsilon L_{\downarrow} \;-\; \varepsilon\sigma T_s^4 \;-\; h_c\,(T_s - T_a) \;-\; q_{\text{cond}} \;=\; 0, \tag{5}$$

where α_{sol} is the solar reflectance, G the global irradiance on the surface, ε the thermal emittance, $L_{\downarrow}$ the incoming long-wave radiation from the sky, $\sigma = 5.67 \times 10^{-8}\,\text{W}/(\text{m}^2\,\text{K}^4)$ the Stefan–Boltzmann constant, T_s the surface temperature, T_a the air temperature, h_c the convective heat transfer coefficient and q_{cond} the heat conducted into the construction.

Table 11: Assumptions for the energy balance. Every value needs a source.

Quantity	**Value used**	**Justification / source**
G	[TODO:] W/m^2	[TODO: MeteoSwiss, clear-sky summer day]
T_a	[TODO:] °C	[TODO:]
h_c	[TODO:] $\text{W}/(\text{m}^2\,\text{K})$	[TODO: low wind speed in a street canyon]
$L_{\downarrow}$	[TODO:] W/m^2	[TODO: clear-sky estimate; state the correlation used]
α_{sol} reference	[TODO:]	[TODO: aged bitumen / plaster]
α_{sol} concept	[TODO:]	[TODO:]

Result and meaning. **[TODO: Solve Eq. (5) for T_s for the reference and for the concept, report both, and state explicitly what the difference does and does *not* prove (see the note on KPI 3 in Section 2.7).]**

3.2.6 Engineering check 2: [TODO: title]

[TODO: Options that are realistic within the week: (a) a simple outdoor measurement — identical plates with different surfaces, logged with a thermometer or a temperature sensor over a few hours; (b) a transient 1-D model of the build-up over a diurnal cycle; (c) a water balance — litres of rainwater per square metre per hot day against the latent heat of vaporisation of 2.45 MJ/kg; (d) a material and cost comparison. Report method, raw data, result and uncertainty.]

3.3 Concept Evaluation & Iteration

Table 12: Decision matrix. Ratings 1–5, weights from Section 2.7. Every rating must be justified in the text.

KPI	**Weight**	**Concept A**	**Concept B**	**Evidence used**
1 – Solar reflectance	[TODO:]	[TODO:]	[TODO:]	[TODO:]
2 – Thermal emittance	[TODO:]	[TODO:]	[TODO:]	[TODO:]
3 – Peak surface temperature	[TODO:]	[TODO:]	[TODO:]	[TODO:]
4 – Heat into the building	[TODO:]	[TODO:]	[TODO:]	[TODO:]
5 – Night radiative cooling	[TODO:]	[TODO:]	[TODO:]	[TODO:]
6 – Potable water use	[TODO:]	[TODO:]	[TODO:]	[TODO:]
7 – Durability of the optics	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Weighted total		[TODO:]	[TODO:]	

3.3.1 Review and feedback

[TODO: World café peer feedback, instructor feedback, and AI used as a critical reviewer — including where the team disagreed with the AI and why. This belongs in Appendix A as well.]

3.3.2 Selection and Final Concept V2

[TODO: State which concept (or justified hybrid) was selected and describe V2 in detail.]

Table 13: Final output of the creating and evaluating phase: Concept V1a/b → engineering review → evidence → Concept V2.

What changed	Why it changed	Evidence that informed the change
[TODO:]	[TODO:]	[TODO:]
[TODO:]	[TODO:]	[TODO:]

3.3.3 Remaining uncertainties

[TODO: What is still unproven, and what should be tested or developed next.]

4 Sustainability & Life's Principles

4.1 Application of Life's Principles to the final concept

[TODO: Take the principles from Section 2.8 and assess each one honestly: how well does the final concept apply it, and where does it fall short? A table with a short assessment per principle works well. Listing principles without assessing them is explicitly marked down in the rubric.]

4.2 Environmental benefits, limitations and trade-offs

[TODO: Quantify where possible. The trade-offs that are already visible for this challenge:]

- **Summer vs. winter:** a high solar reflectance reduces summer heat gain but also reduces useful passive solar gains in winter. **[TODO: estimate both and give the net balance for the Swiss climate]**
- **Cooling vs. water:** evaporative cooling costs roughly 2.45 MJ per kilogram of water evaporated. **[TODO: convert this into litres per square metre per hot day and check it against the rainwater actually available on site — during a heat wave it will not rain, which is precisely the point of the storage function F7]**
- **Local vs. global:** the concept moves heat out of the street canyon; state clearly whether it reduces total energy demand or only redistributes heat.
- **Materials:** **[TODO: pigments, binders, biocides against algae, end of life, separability of the layers, recyclability]**

4.3 Influence of sustainability considerations on the design

[TODO: Name at least one concrete design decision that was changed because of a sustainability consideration — otherwise this section reads as an add-on.]

5 Discussion & Conclusion

5.1 Contribution of the biomimicry process

[TODO: Be concrete and honest. Which idea came specifically from the biological model, and which would a conventional engineering approach have produced anyway? A well-argued "this part was conventional" is worth more than an overstated claim.]

5.2 Strengths and weaknesses of the final concept

[TODO:]

5.3 Assessment against the challenge and the KPIs

[TODO: Go back through KPIs 1–7 from Table 8 and state for each one whether it was met, missed, or could not be assessed with the available evidence.]

5.4 Innovative character and limits of the evidence

[TODO: State clearly what has been demonstrated (calculation, measurement, literature) and what remains an assumption.]

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Appendix A – AI Use Statement

Relevant AI-supported activities are documented below. Not every prompt is listed; the focus is on how AI contributed to the work and how important outputs were verified. AI output is not used as evidence for factual claims.

Table A.1: AI use statement.

AI tool	**Used for**	**How the output influenced the project**	**How important output was verified**
AskNature Chat	Discovering biological strategies for functions F1–F8	**[TODO: e.g. generated candidate organisms for further research]**	**[TODO: AskNature profile + peer-reviewed paper]**
[TODO: LLM assistant]	**[TODO: e.g. structuring the scoping, drafting the LaTeX report template, critical review of the concepts]**	**[TODO: e.g. produced the first draft of the function list, which the team revised]**	**[TODO: e.g. every biological and physical claim checked against the cited primary sources; benchmark values recomputed by hand]**

Appendix B – Optional Supporting Material

[TODO: Additional calculations, extended concept drawings, raw measurement data or prompt excerpts where useful.]